

Chapter 52: Life Cycle Assessment of Green Chemical Processes: A Critical Review of Methods, Evidence, and Decision Practice

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Abstract

Green chemistry lowers the risks and material losses, while LCA verifies whether the improvements are still positive in supply chains and different impact categories. This review combines the principles of ISO-oriented LCA and green chemistry indicators in order to analyze persistent data on pharmaceutical synthesis, CO₂ usage, biomass conversion, and fine chemicals production via catalysis. It is evident that apparent improvements in the atom economy or the yields may be cancelled out by the production of solvents, energy-demanding separations, generation of electricity and hydrogen, catalysts production, feedstocks distribution, and scale-up assumptions. The best studies provide for the definition of the performance-based functional unit, comparable system boundaries, foreground and background inventories, several midpoint impacts, and sensitivity analyses of the most influential parameters. A decision-focused approach is suggested that links the process mass intensity and solvent hazard assessment tools to prospective life-cycle assessment, engineering scale-up, and conditional interpretation. In summary, the chapter indicates that life cycle assessment ought to be incorporated early on and conducted multiple times as the process information becomes more available. Such an approach moves the assessment task from retrospective environmental accounting to route selection and process improvement without burden shifting.

Keywords: *life cycle assessment; green chemistry; process scale-up; prospective LCA; solvents; carbon dioxide utilization; biomass valorization.*

1. Introduction

Green chemistry is assessed using criteria such as waste minimization, atom economy, use of safer solvents, energy efficiency, and catalysis (Anastas & Warner, 1998). Such criteria are very useful in designing molecules and processes; however, they cannot in any way measure upstream electricity, feedstocks production, infrastructures, transportation, or post-consumer phase impacts. The metrics based on mass will continue to be effective while climate, toxicity, water or resources' impacts become worse. The PMI metric is, for instance, an effective operational indicator, as it highlights the materials requirements per unit product; nevertheless, it cannot differentiate between one kilogram of water and one kilogram of scarce, toxic or energy-intensive chemicals (Jiménez-González et al., 2011).

LCA overcomes this drawback by associating the process inputs and outputs with environmental actions in a well-defined product system. Goal and scope definition, life-cycle inventory (LCI), life-cycle impact assessment (LCIA), and interpretation are the stages included in LCA under ISO 14040 and ISO 14044 (International Organization for Standardization [ISO], 2006a, 2006b). However, LCA is still just a model and not a direct measure of sustainability. Different functional units, boundaries, allocations, background databases, impact methods, and time perspectives can influence the outcomes, particularly with new technologies based on chemicals (Cucurachi et al., 2018; Hellweg & Milà i Canals, 2014).

Three questions will be addressed in this chapter: Which methodological choices most strongly determine conclusions for green chemical processes? What recurring hotspots emerge across major application classes? How should LCA be integrated with green chemistry and engineering evidence to

support route decisions? The contribution is a compact, decision-oriented synthesis rather than a numerical meta-analysis: published methods and representative applications are compared to identify robust practices, common failure modes, and research priorities.

2. Review Approach

The structured critical narrative review methodology was employed. In terms of the evidentiary hierarchy, emphasis is laid on ISO standards, European guidance on environmental footprint, existing LCI/LCIA literature, methodological publications regarding chemical process LCA, and selected studies dealing with nascent technologies. References have been considered relevant to the review if they touched upon any of the following aspects: choice of functional unit/boundary design, inventories, scaling up, uncertainties, green metrics, or application to pharmaceutical, CO₂, biomass, or catalytic processes. If only reaction yield, atom economy, E-Factor, or PMI were reported, such references were considered green chemistry.

The four criteria used in interpreting each source included the level of methodological transparency, the quality of the inventory, the level of process maturity, and the relevance of decisions involved. Sources that clearly defined the foreground data, the background database and system model, the allocation rules, the geography, the electricity scenario, and the uncertainty analysis received higher priority. Due to technological and methodological heterogeneity, no numerical meta-analysis was performed.

3. Integrating LCA and Green Chemistry

The two approaches answer different but complementary questions. While green metrics assess the efficiency of reactions and processes, LCA measures the impact of the entire modeled system on the environment. This integration reflects the development of LCA from environmental accounting to decision-making tools and life-cycle sustainability assessment (Guinée et al., 2011). An effective process starts with selecting a similar functional unit – for example, one kilogram of active substance at certain purity or the equivalent service that a chemical can provide. It is important to consider all the phases that are expected to vary among the considered alternatives. Leaving out the production of solvents, hydrogen, feedstocks, catalysts, or waste treatment is a sure way to predetermine the outcome..

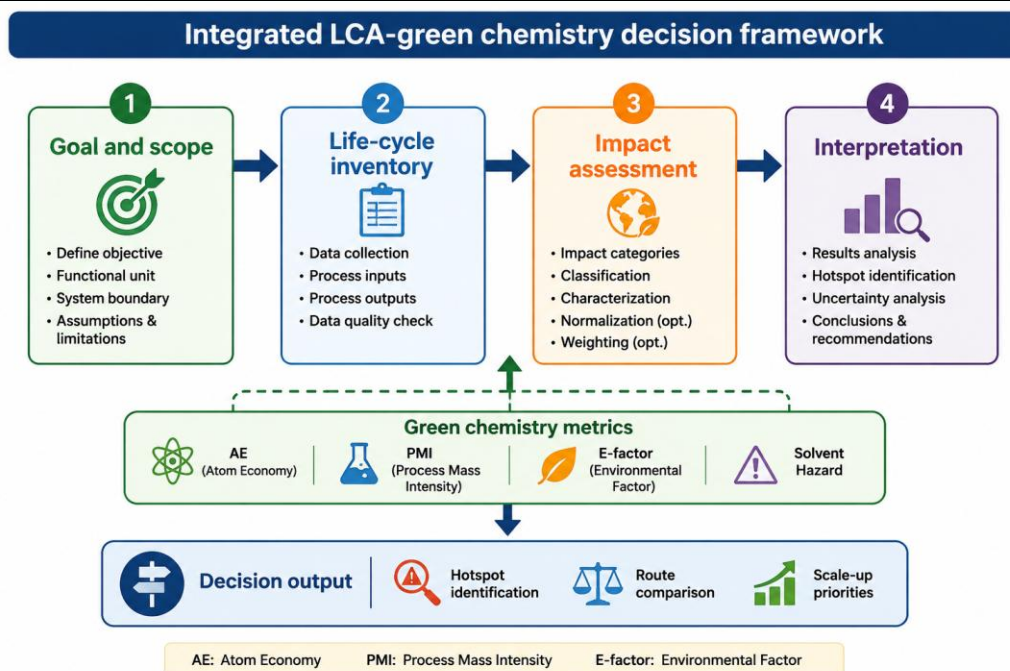


Figure 1. Integrated LCA–green chemistry decision framework. Source: Authors’ original synthesis.

Figure 1 links the four ISO phases with operational indicators. PMI and solvent-hazard tools rapidly locate avoidable material and safety burdens; the LCI then quantifies energy, water, emissions, recovery loops, and upstream production. LCIA translates these flows into multiple indicators, while interpretation tests whether hotspots and comparative rankings survive plausible changes. The value of integration is not a composite “green score,” but a traceable explanation of why one route performs better and which process variable should be changed.

Table 1. Minimum reporting requirements for decision-relevant chemical-process LCA

Element	Minimum requirement	Decision risk if omitted
Functional unit	Quantity, purity, performance, and reference flow	Alternatives may deliver unequal functions
Boundary	Included stages, cut-offs, recycling, and end-of-life	Burden shifting remains hidden
Inventory	Foreground source; database, version, geography, year	Results cannot be reproduced
Multifunctionality	Allocation hierarchy and sensitivity case	Co-product credits can reverse rankings
Scale-up	Measured data plus engineering scenarios	Laboratory energy and solvent use are extrapolated poorly
Uncertainty	Sensitivity, scenarios, and distributions where feasible	False precision obscures fragile conclusions

4. Methodological Choices That Control Results

4.1 Goal, Functional Unit, and Boundary

For there to be any validity in a comparative statement, it must apply to the same kind of function. While a mass functional unit is usually enough for bulk chemicals, some products like medicines or specialty chemicals might need purification, potency, life span, or even performance criteria. Cradle-to-gate cut-off points are useful in the development of routes when the uses and fates are the same; otherwise, cradle-to-grave cut-off points must be used. The cut-off point must be environmentally relevant and not just because of data availability.

4.2 Inventory, Allocation, and Impact Assessment

Background data sets are often employed along with foreground data in chemical inventory. Both database selection and system model need to be reported because different compositions of markets, geographic areas, and allocation rules are involved. Ecoinvent v3, for instance, offers different system models that cannot be substituted in terms of multifunctional processes (Wernet et al., 2016). First of all, allocation can be avoided using subdivision or system expansion in cases that are justified. When allocation is inevitable, mass-based, energy-based, and economic allocation rules need to be tested in cases where co-products have a substantial impact on outcomes (European Commission, 2021).

LCIA methods must adhere to the decision context and geography. Midpoint indicators are usually more directly linked to inventory flows and are favored for process improvement purposes. For instance, ReCiPe 2016 links a large number of midpoint indicators with three endpoint categories without sacrificing the specific value judgements on cultural and time horizons (Huijbregts et al., 2017). The absolute minimum is climate change together with categories related to the chemistry, including particulate formation, acidification, eutrophication, water use, human toxicity, ecotoxicity, and use of mineral or fossil resources. Endpoint aggregation and weighting will help in communication; nevertheless, the value judgments should not hide any trade-offs. Interpretation of the toxicity categories requires caution due to characterization uncertainties.

4.3 Scale-Up and Uncertainty

Scaling laboratory electricity consumption, solvent ratios, and batch losses to industrial production is typically unjustified. The heat integration of laboratory equipment is typically bad, dilution is high, and utilization is low. In contrast, industrial process involves solvent recovery, continuous process, compression, purification, and utility consumption. An engineering approach helps translate laboratory stoichiometry and yield into feasible unit operations and energy balances in the industry (Piccinno et al., 2016). Literature on ex ante LCA indicates that there is still no consistency in upscaling approaches, which calls for more conservative, expected, and optimized scenarios (Tsoy et al., 2020).

The degree of uncertainty analysis must reflect the nature of the decision. The first step in screening involves one-way sensitivity analyses on yield, concentration, solvent recovery rate, catalyst lifetime, electrical mix, and allocation. In scenario analysis, scenarios are combined in a way that creates an internally consistent future. The Monte Carlo approach is appropriate if probability distribution functions can be justified, although it does not solve problems with structure and data. The threshold at which an alternative is favored must be specified.

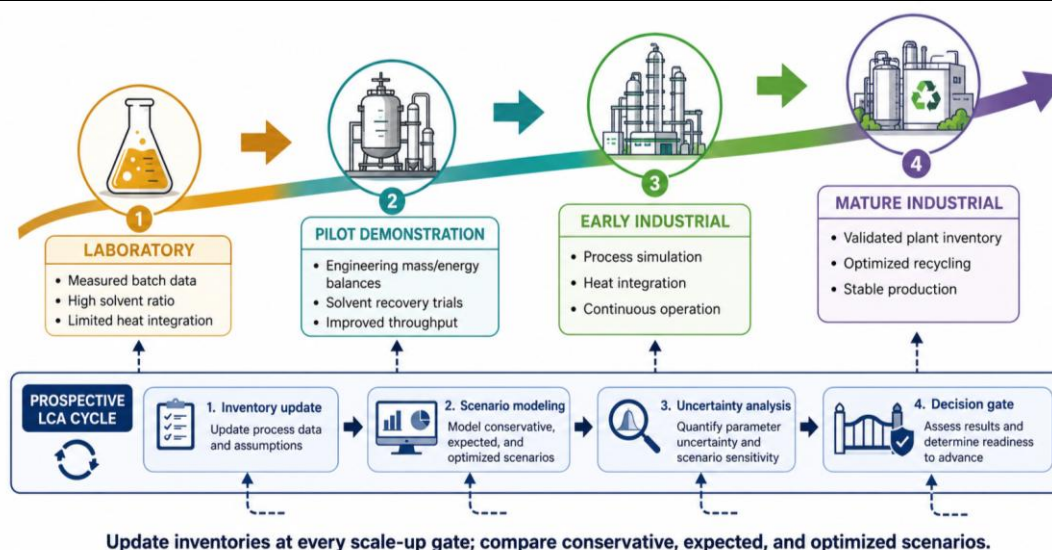


Figure 2. Prospective LCA across the chemical-process scale-up pathway. Source: Authors' original synthesis.

Figure 2 illustrates how prospective LCA evolves across four stages of chemical-process maturity: laboratory, pilot demonstration, early industrial operation, and mature industrial production. Laboratory measurements provide batch-level material and energy data, while pilot studies introduce engineering balances, solvent-recovery trials, and improved throughput. Early industrial models incorporate process simulation, heat integration, and continuous operation, whereas mature assessments rely on validated plant inventories, optimized recycling, and stable production. At every scale-up stage, the LCA cycle updates inventory assumptions, compares conservative, expected, and optimized scenarios, quantifies parameter and scenario uncertainty, and supports a decision gate on technological readiness. This iterative structure prevents premature extrapolation and improves the transparency and reproducibility of environmental assessments (Cespi, 2024).

5. Cross-Case Evidence

[1]. **Table 2.** Recurring hotspots and decision tests across green chemical-process classes

Process class	Recurring hotspots	Priority decision tests
Pharmaceutical and fine-chemical synthesis	Solvents, purification, low yield, reagent manufacture	PMI by step; solvent recovery; route convergence; potency/purity equivalence
CO ₂ conversion	Hydrogen, electricity, compression, separation, short carbon residence	Electricity and H ₂ scenarios; counterfactual CO ₂ source; product lifetime
Biomass valorization	Cultivation or residue credit, moisture, pretreatment, enzymes, allocation	Land-use assumptions; drying and transport; co-product allocation
Catalytic processes	Metal/ligand production, loading, lifetime, leaching, separation	Break-even reuse cycles; catalyst recovery; selectivity–purification trade-off

5.1 Pharmaceutical and Fine-Chemical Synthesis

Solvents and purification are usually common among inventories in pharmaceuticals due to the complexity of the synthesis which involves dilute reactions, repeated extractions, chromatography, and crystallization. This means that PMI will always be relevant in the assessment process, hence its popularity as a tool for route discovery (Jiménez-González et al., 2011). However, LCA provides an additional measure for the intensity and utility of each material in terms of environmental concerns. As such, a slightly less yielding route may prove more efficient than a highly yielding one if it involves elimination of dangerous materials, cryogenics or even solvated isolation processes. The best solution is always to converge steps, telescoping and solvent substitution or recycling, not just optimization of the reaction stoichiometry (Jiménez-González et al., 2013).

5.2 Carbon Dioxide Utilization

Feeding CO₂ into an industrial process will not, in and of itself, result in a climate gain. Outcomes will depend upon the carbon counterfactual, the amount of capture energy, the route of hydrogen production, the composition of electricity, the conversion efficiencies, the separation, the displaced product, and how long carbon will be stored. Methodological guidance highlights the importance of a common system boundary, a sound conventional baseline, and clear handling of avoided emissions in order to avoid double-counting problems (von der Assen et al., 2013; Zimmermann et al., 2020). For fuel or transient products, carbon is generally released; thus, the main climate benefit has to come from lower upstream emissions than the fossil baseline. For durable goods, timing of release may be relevant, although permanence and EoL should be modeled.

5.3 Biomass Valorization

The use of biogenic resources allows lowering the demand for fossil resources while transferring the burden of energy production to agriculture, fertilizers, water, pretreatment and logistics. Waste and residues normally escape from burdens related to cultivation activities, but collection, drying, and alternative use are still valid considerations. The allocation of co-products may be crucial when biorefining processes are under assessment. Comprehensive assessments, in this sense, specify the moisture basis and biogenic carbon balance, distinguish residues from purpose-grown materials, and evaluate land-use and allocation assumptions, comparing the product equivalents.

5.4 Catalysts and Solvents

However, catalytic processes may suffer from increased cost of catalyst preparation and rare metals, from the use of ligands and leaching, and from separation problems. A break-even calculation must find the number of reuse cycles needed to cover the costs of catalyst production and regeneration. Similarly, the fact that the process uses a “bio-based” or “harmless” solvent does not mean it has low environmental impacts; the impacts of energy, recovery, and loss of solvents need to be accounted for. Guides on solvent selection and life-cycle assessment need to be used in combination: hazard ranking ensures the safety of operators and ecosystems, while life-cycle indicators enable comparison of production/recovery burdens (Prat et al., 2016). The E-factor approach continues to be useful for waste characterization, but it needs to be integrated into life-cycle impact analysis (Sheldon, 2017).

6. Decision Framework and Research Priorities

A defensible route-selection process consists of six stages. First, formulate the problem, comparator and performance-based functional unit. Secondly, apply mass balance, PMI and solvents/catalysts hazard checks to spot probable hotspots. Thirdly, create an LCI that distinguishes between observed foreground and calculated/background data. Fourthly, simulate conservative, baseline and optimized scaling up scenarios. Fifthly, present several midpoint impacts and sensitivity analyses for influential variables, allocation, electricity/feedstock scenarios. Sixthly, choose a route only if it is clearly advantageous, or specify the conditions necessary for preference. This way, the analysis remains relevant while adhering to ISO standards.

Research should focus on open, chemistry-specific unit-process inventories; connections between process simulators and LCA tools that are reproducible; consistency in data-quality scores; and uncertainty methodologies that can differentiate between parameter uncertainty, scenario uncertainty, and modeling uncertainty. Machine learning could provide property estimates or inventory estimates for unknown values, but such predictions should come with an applicability domain, validation, and uncertainty estimates. Dynamic models are capable of capturing the dynamics of emissions and the change in background systems that static characterizations fail to capture, hence distorting the decision (Levasseur et al., 2010). Market-mediated change could be captured by consequential models. This added complexity is justified by the decision at hand. Environmental LCA should be analyzed in light of economic, occupational, and social analyses (Bergerson et al., 2020).

6.1 Practical Interpretation and Reporting Protocol

The interpretation process must be initiated prior to model calculation. The research group will formulate a decision statement defining the target audience, alternatives, time period, and consequence of a poor decision. This will define the needed level of precision. Early-stage route evaluation can be based on secondary information and a broad range of scenarios. For public claims regarding comparative analysis, however, primary information needs to be used consistently with critical assessment and comparability control. Data pedigree will need to identify every material influence as measured, calculated, simulated, supplier-reported, data-bank-based, or assumed. This will make information gaps obvious and ensure that a precise numeric value does not imply greater certainty than the input values. It will also help experimental teams to understand which next measurement such as reaction concentration, solvent recovery, catalyst life-time, or energy of separation will most decrease decision uncertainty.

Environmental robustness must be considered at three stages. Parameter robustness concerns how regular fluctuations in yields, recoveries, energy consumption, or transportation affect the results. Scenario robustness considers coherent scenarios like regional electricity systems, alternative biomass supply chains, and either conservative or optimized scale-up options. Method robustness considers structural choices like allocation, substitution credit, background system modeling, life cycle impact assessment approach, and boundary. A process cannot be considered an environmentally preferable option unless its advantage holds across the range of combinations that matter in the choice at hand. Reversal in ranking makes the finding equally useful as long as the threshold of the reversal and supporting evidence are provided. This kind of conditional statement is particularly relevant for novel processes, since optimism about scale-up and credits for avoided products may otherwise overwhelm the modeling.

The recommendations made in the results section must be linked with one of the modeled drivers. An effective results section should include the contribution analysis, the uncertainty range, the scenario matrix, and comparisons with green chemistry indicators in that order. Results related to climate change should not be substituted for overall environmental performance since material toxicity, water stress, resource consumption, and local emissions should be considered when pertinent to both the chemistry and geography. Similarly, reduction of PMI and E-factor should be viewed as efficiency improvements but not as a means of reducing the life cycle impact. Information on software used, database version, system model used, LCIA method used, geography, year of data, cutoff criteria, allocation hierarchy, recycling assumptions, and consideration of biogenic carbon or captured CO₂ must be disclosed. Foreground inventory data and calculation files must be provided in machine readable form when confidentiality allows.

7. Conclusion

LCA strengthens green chemistry through evaluating whether process improvements made locally still maintain their green status at the supply chain and environmental mechanism level. Evidence suggests that these factors keep coming up: solvents and separation; energy and hydrogen; feedstocks production and allocation; catalysts provision; and scale-up assumptions. There is no such thing as a green feedstock, a green catalyst or even a green high yield reaction path. Claims must have comparable functions, complete and clear inventories, multiple impact categories, engineering-based scaling-up, and uncertainty analysis. When used together with PMI and hazard indicators, LCA is a powerful design tool.

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